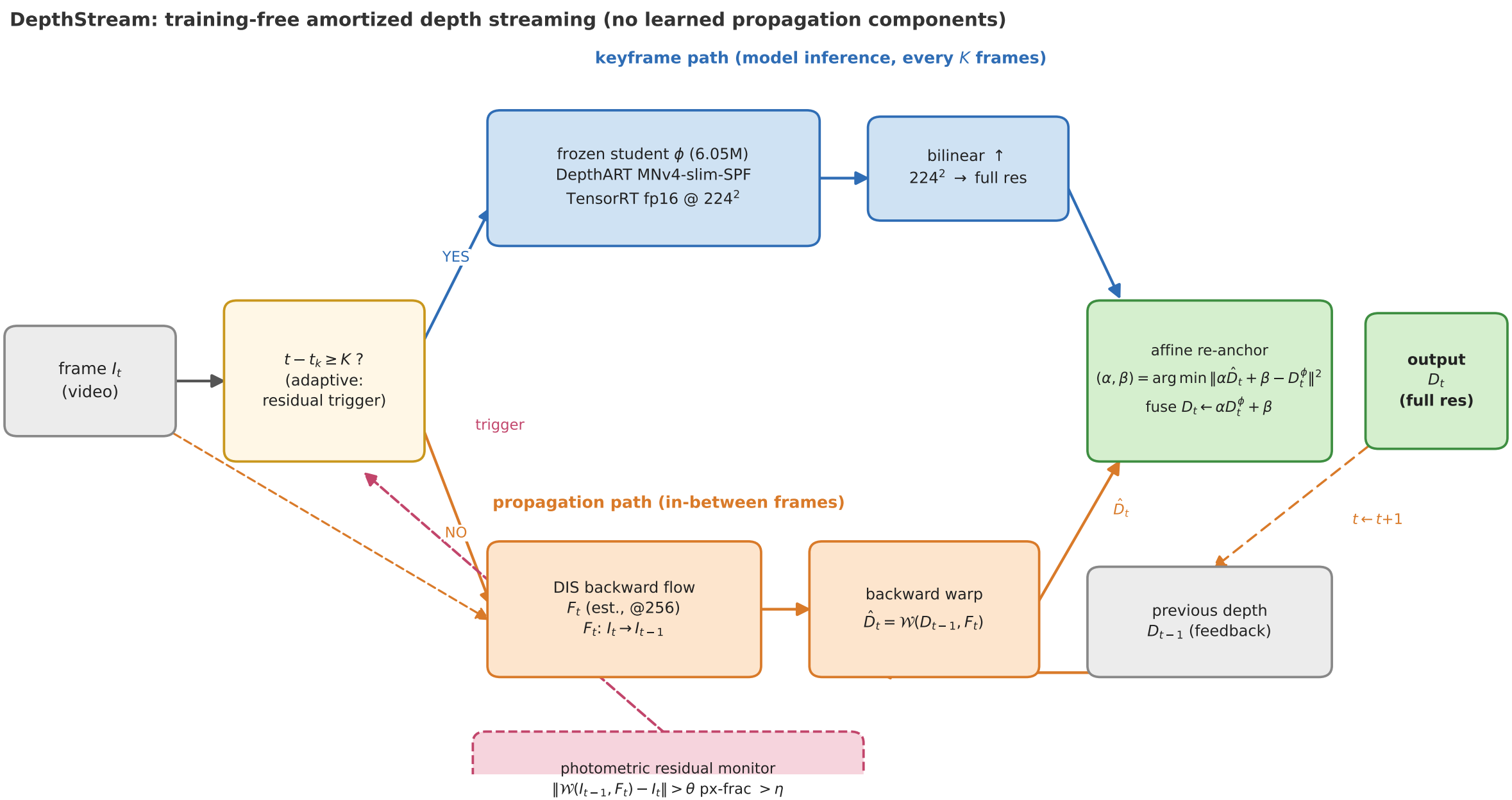
DepthStream: training-free amortized depth streaming (no learned propagation components)
keyframe path (model inference, every K frames)
frame I_t (video)
$t - t_k \geq K$? (adaptive: residual trigger)
YES
frozen student ϕ (6.05M) DepthART MNv4-slim-SPF TensorRT fp16 @ 224^2
bilinear ↑ 224^2 → full res
affine re-anchor $(\alpha, \beta) = \arg\min \|\alpha \hat{D}_t + \beta - D_t^{\phi}\|^2$ fuse $D_t \leftarrow \alpha D_t^{\phi} + \beta$
output D_t (full res)
trigger
propagation path (in-between frames)
NO
$\hat{D}_t$
$t \leftarrow t+1$
DIS backward flow F_t (est., @256) F_t: $I_t \rightarrow I_{t-1}$
backward warp $\hat{D}_t = \mathcal{W}(D_{t-1}, F_t)$
previous depth D_{t-1} (feedback)
photometric residual monitor $\|\mathcal{W}(I_{t-1}, F_t) - I_t\| > \theta$ px-frac $> \eta$
all components are training-free and deployable (TensorRT / ONNX Runtime); color: blue = model path, orange = propagation, green = fusion, pink dashed = adaptive re-anchoring